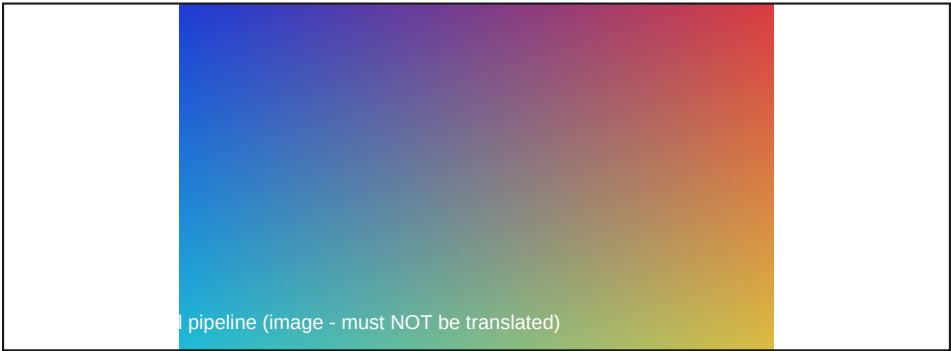


Retrieval-Augmented Generation with Lightweight Re-ranking

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1. Introduction

Retrieval-augmented generation (RAG) has become a standard technique for grounding large language models in external knowledge. Instead of relying solely on parametric memory, the system first retrieves relevant passages from a document store and then conditions the generator on them. This substantially reduces hallucination on knowledge-intensive tasks such as open-domain question answering.



3. Experiments

In this paper we introduce a lightweight re-ranking stage that operates on the top-k candidates returned by a dense retriever. The re-ranker is trained with a contrastive objective on 120,000 question-passage pairs and requires no additional annotation.

Model	Params	EM	F1
Baseline (BM25)	-	31.2	38.4
Dense only	110M	38.7	45.1
Ours (+re-rank)	110M	44.9	51.6

Table 1: Main results (table · must NOT be translated)

2. Method

However, retrieval quality is only half of the story. The generator must also learn to ignore distractors, attribute its claims to the retrieved evidence, and abstain when the evidence is insufficient. Recent work shows that simply concatenating more passages does not monotonically improve accuracy, and can even degrade it when the context window is flooded with irrelevant text.

In this paper we introduce a lightweight re-ranking stage that operates on the top-k candidates returned by a dense retriever. The re-ranker is trained with a contrastive objective on 120,000 question-passage pairs and requires no additional annotation.

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